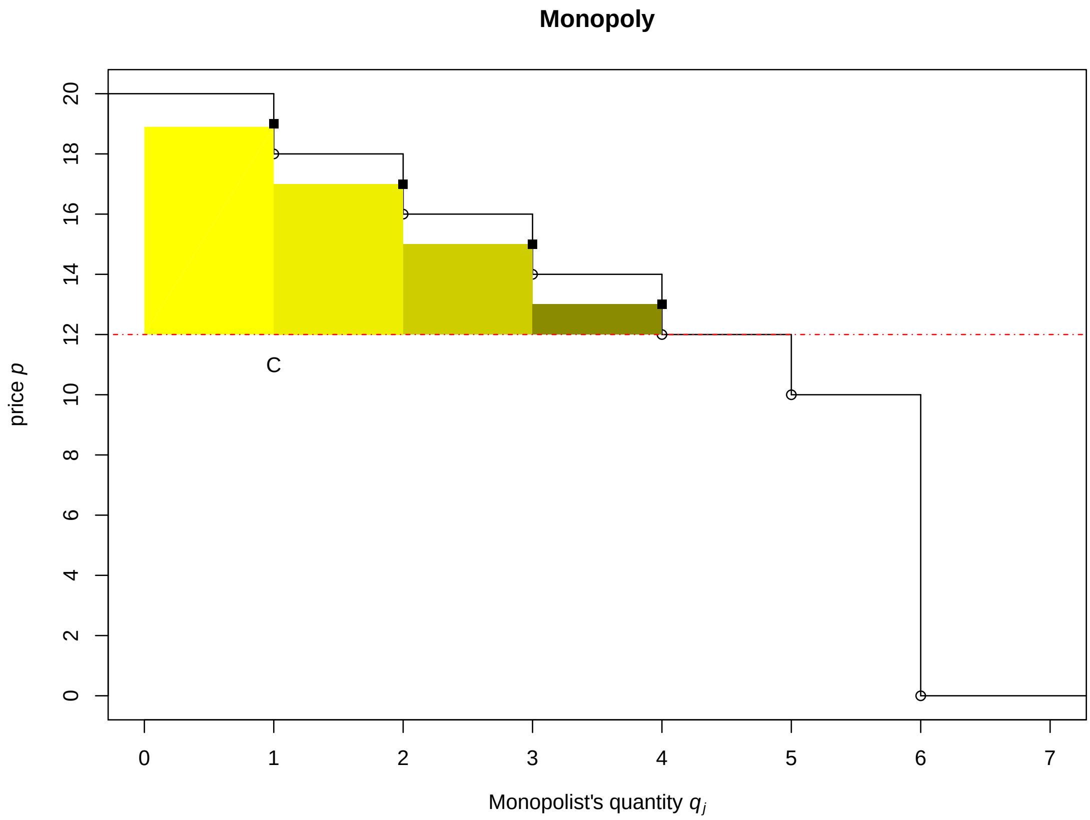
Supplementary Information

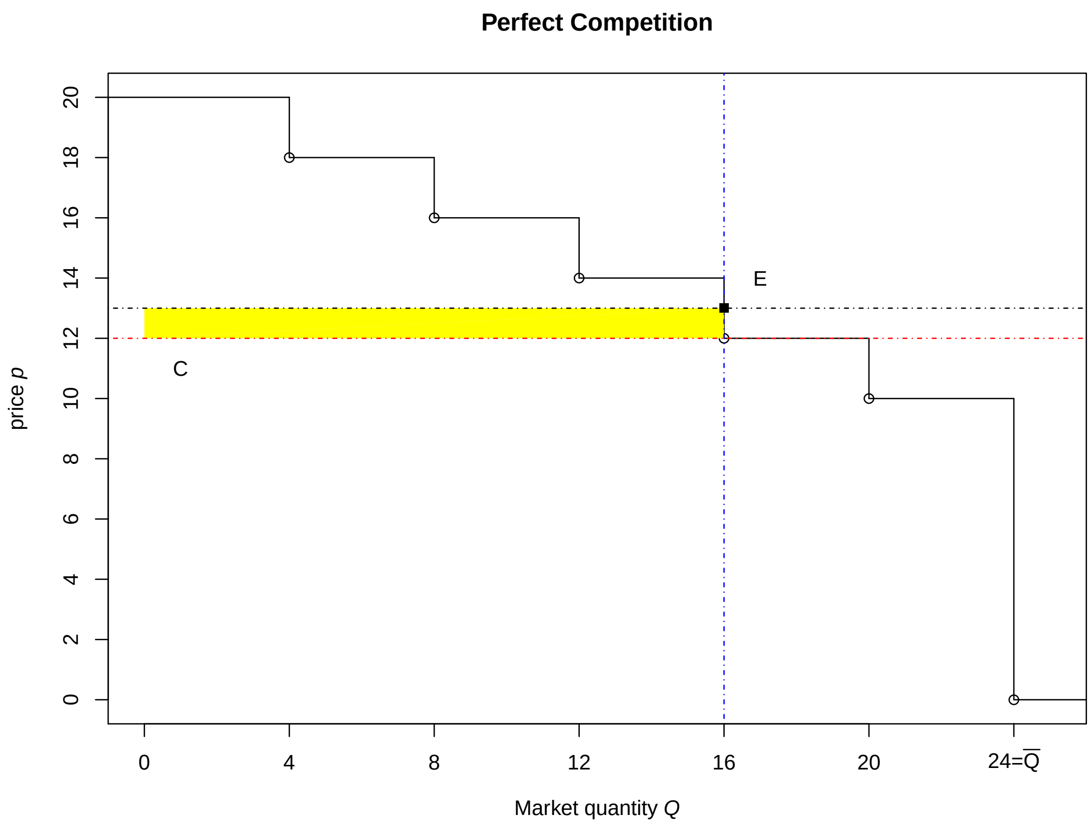
Appendix A - Supplementary Figures

Figure A.1: Demand and Cost functions under Monopoly



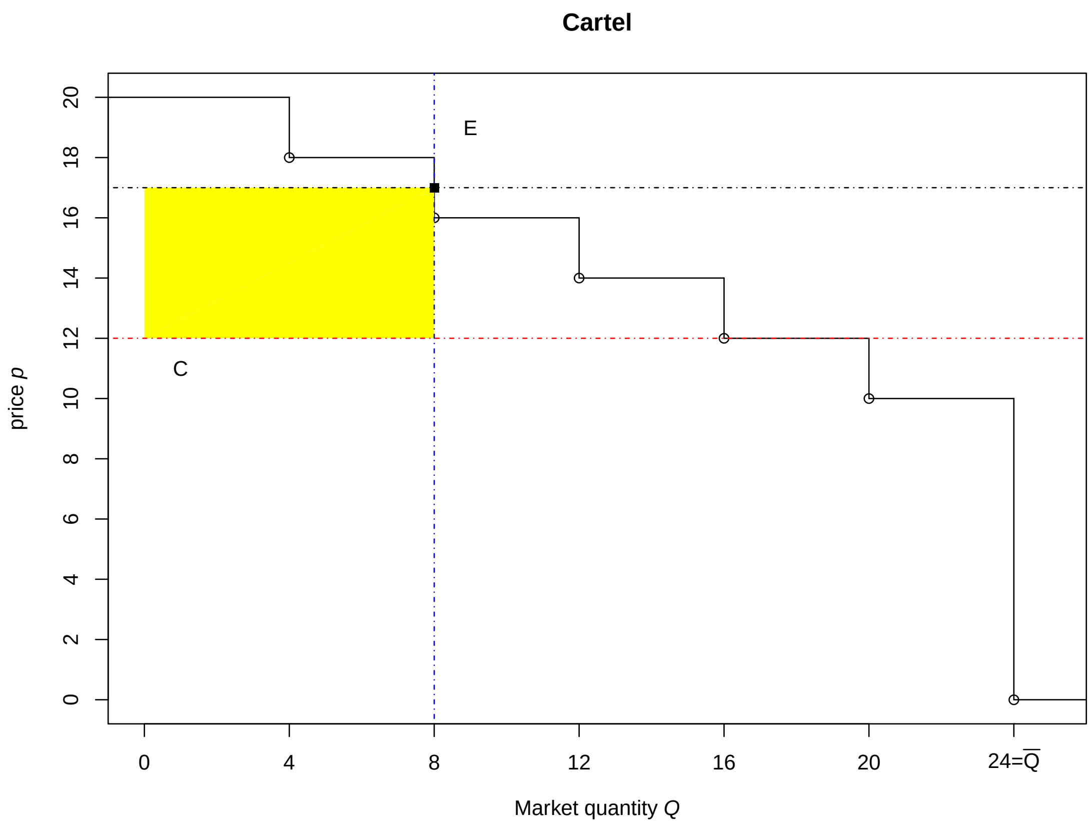
Note: the sum of colored areas represents monopolist j's surplus ($j = 1, 2, 3, 4$).

Figure A.2: Demand and Cost functions under Perfect Competition



Note: the yellow area represents the sellers' surplus.

Figure A.3: Demand and Cost functions under Cartel



Note: the yellow area represents the sellers' surplus.

Figure A.4: Example of a seller's computer screen during the experiment.

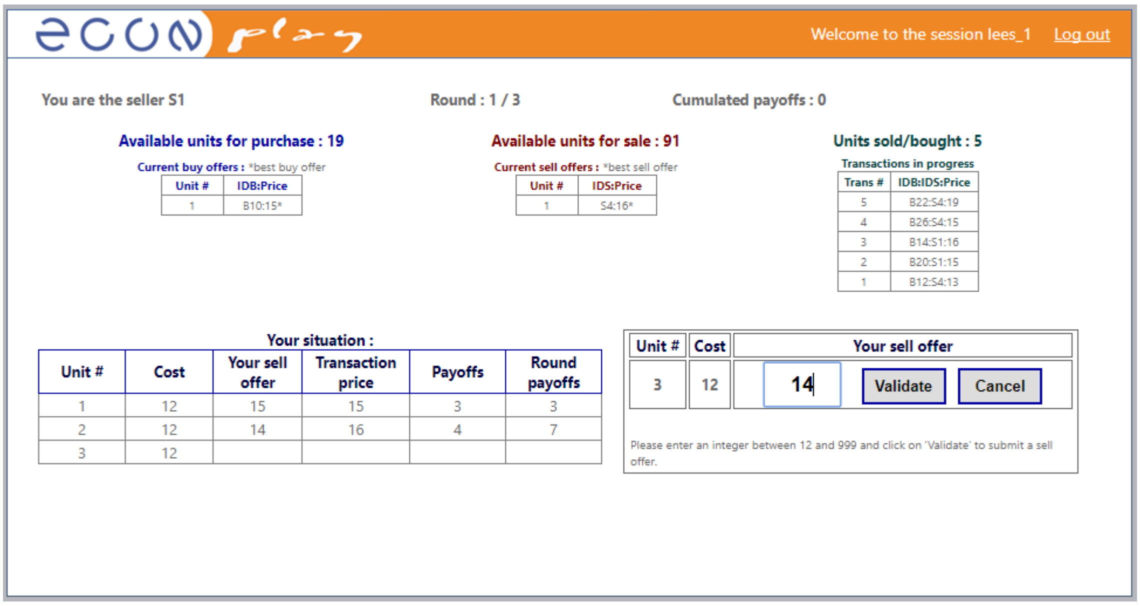
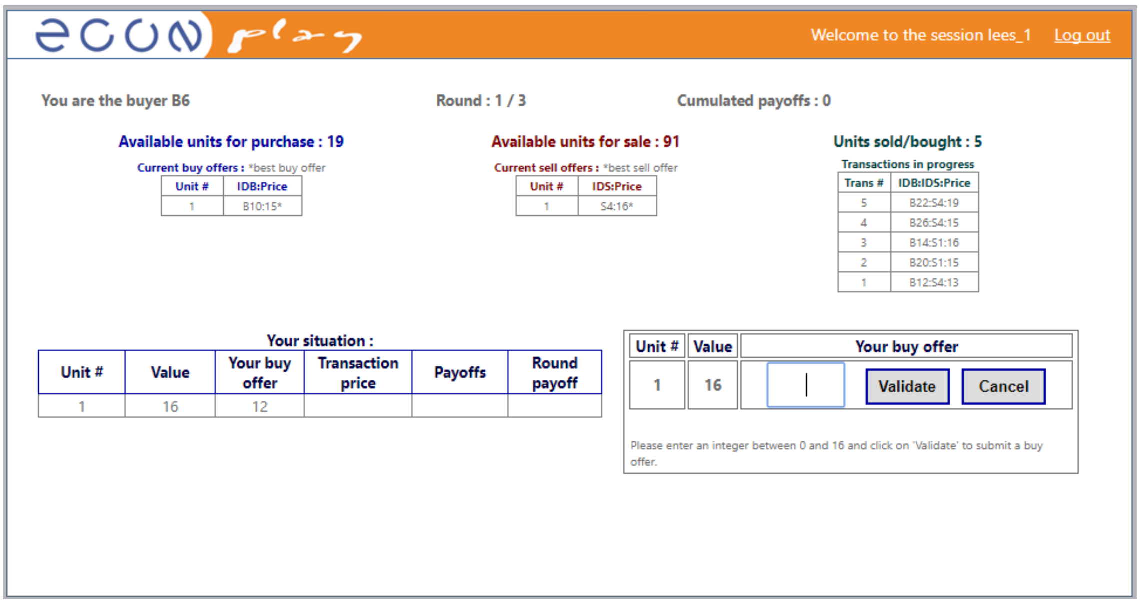


Figure A.5: Example of a buyer's computer screen during the experiment.



Appendix B - Supplementary Tables

Table B.1: Average trading prices for market structures played as first, disentangled by trading period $t = 1, 2, 3$

	p_{Co}	p_{Co-q}	p_{Ca}	p_{Ca-q}	p_{Mo}
$t = 1$	14.49	14.12	14.42	14.97	14.81
$t = 2$	14.10	13.78	14.08	14.30	14.61
$t = 3$	13.50	13.54	13.56	13.76	14.50

Table B.2: Regression models of prices from markets implemented as first in each sequence

	(1) Price	(2) Price	(3) Price$\|t$=1	(4) Price$\|t$=2	(5) Price$\|t$=3
Comp	-0.742^{+}	-0.765^{+}	-0.379	-0.609	-1.121**
	(-1.71)	(-1.70)	(-0.90)	(-1.07)	(-2.88)
Comp-q	-0.945*	-0.990*	-0.797^{+}	-0.886	-1.048**
	(-2.18)	(-2.21)	(-1.91)	(-1.56)	(-2.69)
Cartel	-0.410	-0.459	0.563	-0.619	-1.077**
	(-0.95)	(-1.02)	(1.34)	(-1.09)	(-2.76)
Cartel-q	-0.451	-0.478	0.105	-0.392	-0.865*
	(-1.04)	(-1.07)	(0.25)	(-0.69)	(-2.21)
Period 2		-0.579***			
		(-7.67)			
Period 3		-0.952***			
		(-12.74)			
Constant	14.78***	15.734***	14.94***	14.70***	14.63***
	(94.52)	(92.37)	(83.67)	(70.31)	(99.76)
N (trades)	1542	1542	478	516	548

Notes: Mixed-effect regression models. Random effects at the session, market and seller levels. t-test in parentheses. * *p-value* < 0.05, ** *p-value* < 0.01, *** *p-value* < 0.001.

Table B.3: Fractions of intramarginal sellers and buyers that do not trade, across markets

Market	Sellers $t = 1$	Sellers $t = 2$	Sellers $t = 3$	Buyers $t = 1$	Buyers $t = 2$	Buyers $t = 3$
Comp	4.17%	6.25%	5.56%	20.42%	20.00%	20.97%
Comp-q	2.78%	2.08%	1.39%	21.11%	23.61%	20.42%
Cartel	4.17%	9.03%	4.17%	22.08%	23.75%	20.00%
Cartel-q	8.33%	3.47%	1.39%	26.53%	26.39%	22.22%
Monopoly	2.08%	2.08%	0.69%	33.61%	31.25%	26.67%
Average	4.31%	4.58%	2.64%	24.75%	25.00%	22.06%

Table B.4: Regression models of Monopoly prices after any other market structures

	(1) *Mo* Price	(2) *Mo* Price	(3) *Mo* Price$\|t$=1	(4) *Mo* Price$\|t$=2	(5) *Mo* Price$\|t$=3
Co	-1.000**	-1.016**	-1.269***	-0.856*	-0.961***
	(-3.26)	(-3.27)	(-4.07)	(-2.31)	(-3.34)
Ca	-0.896**	-0.911**	-0.758*	-0.791*	-1.038***
	(-2.92)	(-2.94)	(-2.45)	(-2.14)	(-3.62)
CoCa	-0.693*	-0.711*	-0.769*	-0.698	-0.639*
	(-2.25)	(-2.29)	(-2.46)	(-1.85)	(-2.21)
CaCo	-0.873**	-0.886**	-0.715*	-0.899*	-0.978***
	(-2.84)	(-2.85)	(-2.30)	(-2.43)	(-3.41)
Period 2		-0.219***			
		(-3.29)			
Period 3		-0.403***			
		(-6.14)			
Constant	14.78***	15.01***	14.97***	14.71***	14.62***
	(82.90)	(81.19)	(80.90)	(67.96)	(87.35)
N (trades)	1501	1501	478	495	528

Notes: Mixed-effects regression models. Random effects at the session, market and seller levels. t-test in parentheses. + *p-value* < 0.10, * *p-value* < 0.05, ** *p-value* < 0.01, *** *p-value* < 0.001.

Appendix C - Instructions

[*Translated from French to English; Treatment 1: CoCaMo (see Table 2)*]

The goal of this experiment is to introduce the concepts of monopoly and cartel to students.

General instructions

The experiment consists of 5 consecutive phases. In each of them, there are 4 sellers and 24 buyers of a fictitious good. At the beginning of the experiment the computer will inform you if you are assigned to the role of a seller or a buyer. You will keep the same role throughout the whole experiment.

If you are a seller, you can make profits by selling a unit of the good to a buyer. On your screen, the cost of production of each unit of the good will be displayed. If you sell one unit of at a price P and the production cost of this unit is C, your profit for this unit will be equal to the difference P – C.

If you do not sell any unit of the good, your profit will be zero. You are not obliged to sell. For example, it can be that the price that has been proposed to you for one unit does not cover the production cost. In this case, you will not be able to sell that unit. Furthermore, although the software allows you to sell one unit at the production cost, we recommend you to make at least 1 point of profit for each sold unit.

If you are a buyer, you can buy a unit of the good from a seller. Your valuation for one unit of the good is indicated on your screen: it is the value that you attribute to that unit. Therefore, if your valuation is V and you buy one unit at a price P, your profit will be V – P.

If you do not buy any unit of the good, your profit will be zero. You are not obliged to buy. For example, if the selling price is higher than your valuation, you will not be able to buy the unit of the good. Furthermore, although the software allows you to buy the unit at your valuation, we recommend you to make at least 1 point of profit for each unit that you buy.

Each of the 5 phases that follow consists of 3 trading periods.

Every trading period lasts 2 minutes (120 seconds). In each trading period:

- Each seller can sell several units of the good; each buyer can buy at maximum one unit of the good.
- The production cost is the same for each unit of the good across sellers; the valuation for each good unit can be different for each buyer.
- You can make offers, either to sell or buy, only using integer numbers within 0 and 30.

In each of the 5 phases:

- the unitary production cost for each seller is the same across trading periods;
- the valuation for the same buyer can change across trading periods.

How to exchange: double auction During each trading period, you will have the possibility to submit an ask price (if you are a seller) or a bid (if you are a buyer) according to the same mechanism that we have explained to you during the first tutorial of the course (double auction).

The only difference relative to the first tutorial is that ask prices and bids will be made through a software installed in each of the 28 computers of the laboratory.

Now, we are going to illustrate how to use the software [*sellers and buyers are shown the computer screens reported respectively in Figures A.4 and A.5 of Appendix A*].

If you are a seller: You can submit an ask at any time during the trading period. An ask is entered in the box labeled "Your sell offer" on your screen. The entered ask represents the lowest price that you are willing to accept for a unit of the good.

If you are a buyer: You may submit a bid at any time during the trading period. A bid is entered in the box labeled "Your buy offer" on your screen. The entered bid represents the highest price that you are willing to pay for a unit of the good.

How trades are made: If a seller's entered ask becomes the current lowest ask in the market and it meets or is below the current highest bid, a trade is made between that seller and the buyer holding the current highest bid. Symmetrically, if a buyer's entered bid becomes the current highest bid in the market and it meets or is above the current lowest ask in the market, a trade occurs between that buyer and the seller holding the current lowest ask.

Information shown on your screen: Throughout each trading period, the current highest asks and bids, the number of units already traded (sold/bought) and the number of units still available for sale and for purchase in the market will be shown on all participants' screens (upper part). The bottom part of the screen is your private information. On the right side, it shows the interactive box through which you can submit your asks (if you are a seller) or bids (if you are a buyer). On the left side, it shows your situation in terms of: assigned cost and current lowest ask (if you are a seller), assigned valuation and current highest bid (if you are a buyer), price at which you have traded a unit and profit obtained for that unit.

On the one hand, sellers can sell a given number of units of the good to sell, which varies depending on the phase. On the other hand, each buyer can only buy a maximum of one unit of the good in each period.

Is everything clear? If you have questions, please raise your hand and the experimenter will answer them privately.

[*At the end of the General Instructions, instructions of Phase 1 are shown on each participant's screen.*]

Phase 1 (*Perfect Competition*)

In each of the 3 trading periods:

- Each of the 4 sellers can sell 24 units of the good.
- Each of the 24 buyers can buy at maximum one unit of the good.

[*At the end of Phase 1, instructions of Phase 2 are shown on each participant's screen.*]

Phase 2 (*Perfect Competition with quotas*)

This phase is the same as Phase 1. The only difference is that, in each trading period:

- Each of the 4 sellers can sell at maximum 6 units of the good.

[*At the end of Phase 2, instructions of Phase 3 are shown on each participant's screen.*]

Phase 3 (*Cartel*)

This phase is the same as Phase 1 (each of the 4 sellers can sell at maximum 24 units of the good). The only difference is that, before the start of the phase, the 4 sellers can agree upon the price to apply to all the units to sell in the subsequent trading periods of this phase.

The agreement is implemented through the following rules (keeping anonymity among sellers):

- Each seller writes down on a paper the price that he/she would like to apply to all the units in the following three trading periods of this phase.
- The experimenter collects each price proposal from each seller.
- The experimenter computes the average of the prices that the four sellers have indicated: such average price is the only one that the four sellers should offer in each trading period during this phase.

The experimenter shows to each seller the price proposed by the other three sellers and the average price they should propose.

If they want, sellers can make offers that deviate from such average price. They are not obliged to respect the agreement.

[*At the end of Phase 3, instructions of Phase 4 are shown on each participant's screen.*]

Phase 4 (*Cartel with quotas*)

This phase is the same as Phase 3. The only difference that, in each trading period:

- Each of the 4 sellers can sell at maximum 6 units of the good.

[*At the end of Phase 4, instructions of Phase 5 are shown on each participant's screen.*]

Phase 5 (*Monopoly*)

In each of the 3 trading periods, there are 4 similar markets. The four markets are similar in the sense that, in each of them:

- There are only 1 seller and 6 buyers.
- The seller can sell at maximum 6 units of the good.
- Each of the 6 buyers can buy at maximum one unit of the good.
- The distribution of the valuations of the 6 buyers is the same among markets.